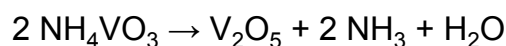


When heated, ammonium metavanadate decomposes according to the reaction below.



Which of the following statements correctly describes the oxidation state of vanadium during the reaction?

- ☐ A. The oxidation state of vanadium changes from +6 to +5. [26%]
- ☐ B. The oxidation state of vanadium changes from -1 to 0. [6%]
- ☐ C. The oxidation state of vanadium is 0 and does not change during the reaction. [6%]
- ✓ ☒ D. The oxidation state of vanadium is +5 and does not change during the reaction. [60%]

Correct

59% answered correctly

Explanation:

Oxidation state of vanadium during the reaction

	Reactant				Product	
Oxidation number (single atom):	-3	+1	+5	-2	+5	-2
Compound formula:	NH_4VO_3				V_2O_5	
Totals (column):	-3	+4	+5	-6	+10	-10
Sum of the totals row must equal the net charge	$-3 + 4 + 5 - 6 = 0$				$+10 - 10 = 0$	

The oxidation state of vanadium is +5 and does not change during the reaction.
© UWorld

Oxidation numbers account for the gain or loss of electrons by an atom relative to the number of electrons present in its elemental state. Oxidation numbers can be assessed from the compound formula using [oxidation state rules](#). By this method, atoms with consistent oxidation states (ie, those that rarely vary) are used to determine oxidation states for atoms that have variable states (eg, [transition metals](#)).

Oxidation numbers differ from the [formal charge](#) of an atom in that oxidation numbers account for electron changes due to [oxidation and reduction](#) whereas formal charge accounts for distribution of charge among atoms. However, in any molecule the **sum of the oxidation states** of every atom in the structure must equal the **net charge** of the compound formula.

Using the **oxidation state rules** for the formulas NH_4VO_3 and V_2O_5 shows that vanadium has an oxidation state of +5 in both compounds. Because neither NH_4VO_3 nor V_2O_5 has a net charge, the oxidation states of the atoms in each compound must sum to zero. As such, the oxidation state of V in NH_4VO_3 is evaluated as:

$$\text{NH}_4\text{VO}_3 = 0 \text{ (net charge)}$$

$$1(-3) + 4(+1) + 1(\text{V}) + 3(-2) = 0$$

$$-3 + 4 + \text{V} - 6 = 0$$

$$V = +5$$

Similarly, the oxidation state of V in V_2O_5 is evaluated as:

$$V_2O_5 = 0 \text{ (net charge)}$$

$$2(V) + 5(-2) = 0$$

$$2V = +10$$

$$V = +5$$

Therefore, the oxidation state of **vanadium is +5** and **does not change** during the reaction.

(Choice A) NH_4VO_3 consists of NH_4^+ and VO_3^- ions. Consequently, the V atom in VO_3^- must have an oxidation state of +5 so that the sum of the oxidation states for the ion is -1 .

$$VO_3^- = -1 \text{ (net charge)}$$

$$1(V) + 3(-2) = -1$$

$$V - 6 = -1$$

$$V = +5$$

(Choice B) The VO_3^- ion has a net charge of -1 and the V_2O_5 molecule has a net charge of 0, but these charges are not equivalent to the oxidation state of V.

(Choice C) The V atoms have a formal charge of zero in all compounds, but the formal charge is not the same as the oxidation state of the atoms.

Educational objective:

Using oxidation state rules, the oxidation number of each atom in a compound formula may be determined on the basis that the sum of the oxidation states of each atom must equal the net charge of the compound or ion.

Time spent: 0 sec

Subject:
General Chemistry

QID: 403958

System:
Interactions of Chemical Substances